QUANTUMFLOOR

Computational Research Report

Job ID: c751468b

1. Executive Summary

This report details the execution of the QuantumFloor diagnostic pipeline on dataset 'label'. The system evaluated multiple classical baselines against quantum algorithms to determine the optimal strategy.

2. Dataset Intelligence

Modality:	TABULAR
Total Rows:	40
Unique Rows:	40
Duplicate Rows:	0
Total Columns:	5
Missing Values:	0.00%
Leakage Risk Detected:	0

3. Data Preparation

Train/Test Split:	80/20 Stratified
Preprocessing:	Pipeline (Imputation + Scaling)
Leakage Protection:	Enforced

4. Computational Plan

Original Features:	5
Quantum Embedding Dimension:	2
Qubit Budget:	2
Classical Training Size:	32
Quantum Subset Size:	32

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5. Quantum Execution

Execution Backend: SIMULATED EXECUTION

Ansatz: AngleEmbedding + IsingZZ

Qubits: 2

Circuit Depth: 2

Total Operations: 2

6. Classical vs Quantum Baselines

Model	Type	AUROC	F1	Spec@95	Notes
full_feature_classical	Classical	0.4375	0.5000	0.0000	
logreg	Classical	0.3750	0.3333	0.0000	
svm_rbf	Classical	0.3750	0.0000	0.0000	
random_forest	Classical	0.4375	0.4444	0.2500	
qsvm	Quantum	0.5000	0.2857	0.2500	
pqk	Quantum	0.3125	0.0000	0.0000	
vqc	Quantum	0.6250	0.4444	0.2500	

7. Comparison & Explainability

Best Classical AUROC: 0.4375

Best Quantum AUROC: 0.6250

Delta: 0.1875

Explainability Status: available

PC2: 0.2262

PC1: 0.1607

8. Final Verdict

Best Overall Model: vqc

Quantum Advantage: Yes